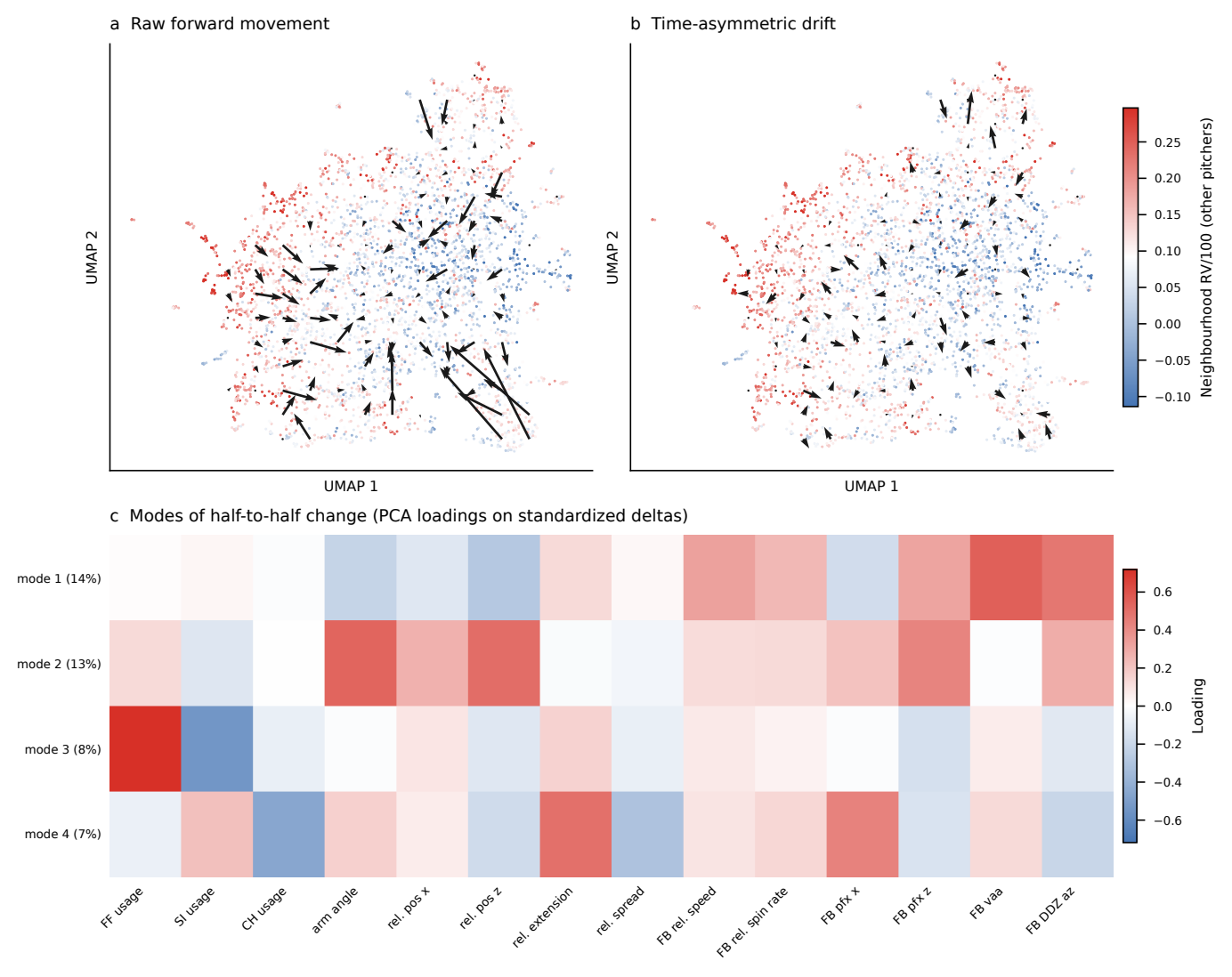

a Raw forward movement
b Time-asymmetric drift
UMAP 2
UMAP 1
Neighbourhood RV/100 (other pitchers)
0.25
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
c Modes of half-to-half change (PCA loadings on standardized deltas)
mode 1 (14%)
mode 2 (13%)
mode 3 (8%)
mode 4 (7%)
Loading
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
FF usage
SI usage
CH usage
arm angle
rel. pos x
rel. pos z
rel. extension
rel. spread
FB rel. speed
FB rel. spin rate
FB pfx x
FB pfx z
FB vaa
FB DDZ az